

# Finding More, Fusing Less: A Pre-Registered Locator Bake-off for AI Agents

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**Companion report to** *Reading More, Finding Less: A Pre-Registered Anatomy of Progressive Disclosure for AI Agents* (doi:10.5281/zenodo.21960138). This study is registered as conjecture c36 in the same evidence program; predictions were frozen before any test question existed, adjudication was mechanical, and the verdict is **revised** — one prediction failed, and the failure retired a shipped default. Both outcomes are reported at the same prominence.

## Abstract

The parent study attributed a curated disclosure policy’s failure to mis-routing by its index, but left the *winning* arm’s own miss ceiling — associative search failing to reach the gold document on 19.6–25% of questions — unattributed. This companion tests the constructive converse: whether that ceiling is reducible by ranked lexical retrieval, under pre-registered predictions frozen before any test question existed. On the parent’s frozen 141-document public corpus, with a fresh 115-question set generated under mechanically enforced gates, four locator arms ran behind an identical  $k=8$  tool contract. BM25 with per-language analyzers located the gold document on 93.0% of questions against hardened exact search’s 71.3% (+21.7 pp; McNemar  $p < 1e-5$ ) — the miss ceiling was substantially a ranking problem. The primary registered prediction passed: the fused BM25+exact locator beat exact search by +13.9 pp ( $p = 0.00014$ ) against a +5.0 pp threshold. But a second frozen prediction — that reciprocal rank fusion never costs localization against its own best leg — **failed**: the fusion trailed BM25-alone by 7.8 pp ( $p = 0.0225$ ), and the pre-registered ablation rule retired the shipped fusion default automatically. An off-the-shelf SQLite FTS5 control (89.6%) was not statistically distinguishable from the custom implementation, assigning the gain to BM25 as such rather than to our code. Downstream answer accuracy moved +5.2 pp, decomposing into +7.5 pp from improved localization partly offset by -2.3 pp of conditional accuracy: reaching the right document remains the binding stage, replicating the parent’s anatomy on a third, blind question set. The findings were independently validated by an isolated re-computation session that reproduced 21 of 22 claimed numbers exactly and confirmed the verdict from the raw records. The full harness, question set, run records, and adjudicator ship for re-running.

## 1. Why this study exists

The parent study [1] built a per-question localization instrument to show *why* a curated index-then-hydrate policy loses to grep-then-read: the index mis-routes, and answers issued without the gold document are almost always wrong. That anatomy left its winner unexamined. Search reached the gold document on only 80.4% (private corpus) and 75.0% (public corpus) of seen-set questions, and nothing in either study decomposed the misses. A post-hoc decomposition of the private-corpus misses suggested the dominant profile was same-language morphology and ranking — matches returned in file order, selection left to the policy — rather than vocabulary mismatch.

That suggestion is the kind of claim this program does not leave as interpretation. c36 registered it: if the ceiling is a ranking problem, a ranked lexical locator should raise localization at bounded cost, and the registration staked numeric predictions on arm-vs-arm deltas before a single test question existed.

**Contributions.** (i) A pre-registered, mechanically adjudicated comparison of locator compositions behind an identical agent tool contract — a design that, per a live-verified literature sweep (§6), no published empirical study occupies. (ii) The pre-registered **retirement of a shipped default**: reciprocal rank fusion, deployed as the framework’s standard locator, failed its frozen never-costs prediction and was removed by rule, not by judgment. (iii) A third replication, on fresh blind questions, of the program’s central anatomy: accuracy conditional on reaching the gold document is stable (~50–54%) across locator quality, so localization gains pass through to answers nearly one-for-one. (iv) A question-generation protocol whose constraints are enforced by mechanical gates rather than instructions — and whose gate-rejection profile itself replicates the parent’s finding that instructed constraints are not constraints.

## 2. Registration

The conjecture (c36) and its registration were committed 2026-08-16 with five frozen predictions and a mechanical verdict mapping, **before the question set existed** — a deliberate tightening of the program’s usual order, so no prediction could be informed even by the questions’ surface difficulty. All predictions are arm-vs-arm deltas on one fresh confirmatory set, making them robust to difficulty shifts between question sets. Localization means the gold document appears in the arm’s  $k=8$  hit list.

| #  | Frozen prediction                           | Threshold                                         |
|----|---------------------------------------------|---------------------------------------------------|
| P1 | fused (L2) beats exact (L0) on localization | $\geq +5.0$ pp                                    |
| P2 | fused never trails BM25-alone (L1)          | $\Delta \geq 0$ pp                                |
| P3 | Spanish-gold subset gain                    | $\geq +8.0$ pp, adjudicated only at $n \geq 25$   |
| P4 | cost bounds                                 | build $\leq 15,000$ ms; added query $\leq 500$ ms |
| P5 | downstream accuracy does not regress        | $\Delta \geq 0$ pp                                |

Verdict mapping, frozen: *refuted* if  $L2-L0 \leq 0$ ; *supported* if P1, P2, P4 pass, P3 passes or is underpowered, and P5 passes; *revised* otherwise. Interpretation cannot change the outcome.

### 3. Design

**Corpus.** The parent study’s frozen public snapshot: 141 markdown documents (1,162,998 bytes), every file hash re-verified against the committed manifest at each run setup; any mismatch aborts.

**Arms.** Four locators behind one pinned contract ( $k=8$  hits, snippets  $\leq 240$  chars, identical read primitives, same answerer, prompts, scoring and adjudicator — the locator is the only variable):

- **L0** — hardened exact search (ripgrep-class term matching with deterministic heuristic ranking); the baseline, and the direct analogue of the parent’s winning policy’s search tool.
- **L1** — Okapi BM25 ( $k1=1.2$ ,  $b=0.75$ ) over an in-memory index with per-language (Spanish/English) Snowball analyzers and field boosts.
- **L2** — reciprocal rank fusion ( $k=60$ ) of L0 and L1’s rank lists; the framework’s shipped default and the primary registered arm.
- **L4** — SQLite FTS5 (porter unicode61, BM25 ranking) behind the same contract: an off-the-shelf external control that separates “BM25 the idea” from our implementation. Descriptive only; staked by no prediction.

A fifth registered arm (cross-lingual dense retrieval) was not built or run; its pre-registered activation trigger had not fired.

**Questions.** A fresh set, because the seen sets’ results are published and a prediction frozen against a known baseline is not blind. The generator reused the parent study’s frozen prompt and model verbatim, but every constraint became a mechanical acceptance gate with reject-and-regenerate (max 3 attempts per document): answer verbatim in the body; **3–15 words counted by the validator, not instructed to the model**; not recoverable from the question itself; present in at most two corpus documents; not degenerate. Of 141 documents, 115 yielded an accepted question; 26 exhausted their attempts and are excluded and listed. The gates fired 227 times (length 75, multi-document 59, verbatim 46, degenerate 45, in-question 2) — the parent measured a 30% violation rate when the same length constraint was merely instructed, and this profile is that finding reproduced from the enforcement side. The salted-split rule assigned all 115 to the confirmatory set (the registered 120+10 could not be met from the yield; the shortfall and its disposition were recorded before any arm ran, and no rule was altered after the yield was known).

**Downstream policy (P5).** One locate per question per arm; the answerer (a small fixed model) receives the full bodies of the arm’s top-8 documents in rank order (truncated at 6,000 chars with visible markers) and answers in a fixed format. One model call per question per arm — a bounded hydrate-all policy that makes the locator the only varying input. Consequence, stated plainly: absolute accuracies here are not comparable to the parent’s agentic-loop numbers; only the  $L2-L0$  delta is interpreted, which is all P5 stakes.

**Scoring.** A word-boundary containment rule adjudicates (closing a substring false-positive class the parent disclosed); the parent’s exact rule is computed alongside descriptively. The

two agree within 0.9 pp everywhere; no outcome depends on the choice.

## 4. Results

### 4.1 Localization (n = 115, gold in top-k)

| Arm             | @8           | @3           | @1           |
|-----------------|--------------|--------------|--------------|
| L0 exact        | 71.3%        | 60.9%        | 43.5%        |
| <b>L1 BM25</b>  | <b>93.0%</b> | <b>84.3%</b> | <b>65.2%</b> |
| L2 fused        | 85.2%        | 73.0%        | 53.9%        |
| L4 FTS5 control | 89.6%        | 76.5%        | 55.7%        |

Paired-design statistics (exact McNemar, computed in independent validation and adopted here as descriptive): L1 vs L0, 28-vs-3 discordant pairs,  $p < 1e-5$ . L2 vs L0, 17-vs-1,  $p = 0.00014$ . L2 vs L1, 2-vs-11,  $p = 0.0225$ .

### 4.2 Adjudication

| #  | Measured                                                        | Outcome                           |
|----|-----------------------------------------------------------------|-----------------------------------|
| P1 | +13.9 pp (85.2 vs 71.3)                                         | <b>pass</b> (threshold +5.0)      |
| P2 | -7.8 pp (85.2 vs 93.0)                                          | <b>fail</b>                       |
| P3 | subset n = 0                                                    | underpowered; not adjudicated     |
| P4 | 193.7 ms build; +147.4 ms/query (difference of per-arm medians) | <b>pass</b> (bounds 15,000 / 500) |
| P5 | +5.2 pp (43.5 vs 38.3)                                          | <b>pass</b>                       |

**Verdict: revised** — the primary prediction passed and an adjudicated secondary failed. The verdict was computed by the frozen analyzer and independently re-derived from the raw records (§7).

### 4.3 The anatomy, a third time

Accuracy conditional on reaching the gold document: 53.7% (L0, n=82) and 50.0% (L2, n=98). Accuracy otherwise: 0.0% (n=33) and 5.9% (n=17). The +5.2 pp downstream gain decomposes into +7.5 pp from localization, partly offset by -2.3 pp of conditional accuracy. Localization is where the outcome is decided; comprehension of a reached document is roughly constant across locator quality. This is the parent’s central mechanism reproduced on a blind question set with a different policy shape.

## 5. The fusion retirement

The interesting failure. RRF was adopted on the standard warrant — fusion “consistently yields better results than any individual system” [2] — and the registration staked exactly that: P2, fusion never costs localization against its own best leg. It cost 7.8 points.

The literature, swept and live-verified after adjudication, shows the *phenomenon* is documented: RRF is parameter-sensitive and beatable by learned or calibrated alternatives [3]; a “weakest link” path can drag a hybrid blend below its best component [4]; and an industry deployment found fusion’s recall gains neutralized under fixed depth and latency contracts [5]. What those studies do not contain is this design: a *prior, frozen* prediction that fusion would not cost localization, mechanically adjudicated, with the shipped default retired by rule on the result. c36’s claim is the pre-registered retirement, not the discovery that fusion can fail.

Mechanistically, the configuration matches [4]’s weakest-link condition in its most extreme form: one leg (BM25) is a calibrated ranking; the other (exact) returns matches in file order with heuristic scoring — closer to an unranked set than a ranking. RRF assumes rank positions carry comparable information in both lists; here the exact leg’s positions are substantially noise, and fusing noise into a good ranking degrades it. Two concessions bind: the RRF constant was fixed at  $k=60$  and never swept, so parameter sensitivity [3] is a live confound; and no published work isolates the specific ranked-with-unranked fusion condition, so the mechanism reading, while consistent with the data, is not itself adjudicated.

The external control disciplines the success story the same way the registry disciplined the failure: FTS5 at 89.6% is not statistically distinguishable from our 93.0% implementation on this n. BM25 as such carries the gain; the custom per-language implementation is not yet shown to add anything. That is the honest sentence, and it is cheaper to write now than to retract later.

## 6. Related work

Positions here follow a live-verification discipline: every citation was opened at source during an isolated literature session, one load-bearing sentence quoted verbatim and checked, before entering this section. Author lists are omitted where not independently verified; identifiers are the citation.

**Fusion.** [2] (SIGIR 2009) is the confronted warrant. [3] (TOIS 2023) pre-empts “RRF can lose” and supplies the parameter-sensitivity concession. [4] (arXiv:2508.01405) names the weakest-link mechanism across 11 datasets. [5] (arXiv:2603.02153) is the nearest production analogue of the failure.

**Lexical retrieval for agents.** [6] (arXiv:2605.15184) and [7] (arXiv:2608.01507) bracket the grep-vs-retrieval question on agent substrates, the latter concluding for indexable read-only corpora that retrieval is “the stronger and cheaper option” — c36’s direction on a different substrate. [8] (arXiv:2608.05886) bounds our finding sharply: BM25 retrieval *below a precision threshold degrades* a downstream agent — so +21.7 pp is a localization-recall result, not a general “BM25 helps agents” claim. [12] (BEIR, NeurIPS 2021) grounds BM25’s robustness and explains why the FTS5 control lands where it does.

**Localization binds downstream.** [10] (arXiv:2503.09089) and [11] (ToolRet, ACL 2025) estab-

lish the reach-to-outcome chain on code and tool substrates; our §4.3 is a third convergence, contributed as a decomposition rather than a discovery. [14] (RAGGED, ICML 2025) is the strongest counter-framing — reader noise-robustness, not retrieval, as the key determinant of RAG stability. The tension is resolved by design class: RAGGED’s readers integrate many retrieved passages under noise; our policy hydrates a small fixed set, so when the gold document is absent nothing downstream can recover (0–6% accuracy), which is precisely the regime where localization binds. Outside that regime RAGGED’s assignment may well dominate; we claim nothing there.

**Method.** [9] (arXiv:2606.22417) is prior art on single-variable within-harness retrieval ablation with a localization-to-resolution chain; c36’s methodological addition over it is the frozen prior prediction and the rule-bound retirement. [13] (arXiv:2606.11217, ICML 2026 position) catalogs the researcher degrees of freedom this design freezes — and the sweep found *no empirical pre-registered retrieval-composition study for agents*, which is the gap this report occupies.

## 7. Independent validation

Before this report was written, the frozen artifacts and complete raw records were handed to an isolated validation session (no web, no repository access, attachments only) instructed to distrust every stated number and recompute from records. It reproduced 21 of 22 claimed numbers exactly — the exception being a  $\pm 0.1$ –0.8 ms estimator ambiguity in one latency figure against a 500 ms bound — applied the frozen verdict mapping to its own values, and confirmed **revised** follows. Its findings were adjudicated in a dated disposition and are incorporated above: the McNemar statistics (§4.1), the FTS5 non-significance (§5), the P5 decomposition (§4.3), and the limitations below. A parallel isolated session performed the literature verification of §6 (13 works, all live-verified, zero failures).

The validation’s declared boundaries are part of the record: it could not verify the corpus snapshot hashes, the locator build, model identities, or commit ordering from the attachments alone. The public bundle exists to close those boundaries: corpus manifest, pinned harness, run records, and adjudicator ship for external re-running.

## 8. Limitations

1. **One corpus, one harness family, one contract.** All claims are scoped to a frontmattered operational documentation corpus behind a  $k=8$  contract; [6] shows harness and tool-presentation effects can be large independently of the retriever.
2. **The registered measure set was not fully delivered.** Marginal tokens were recorded only as prompt characters; index peak RSS was not captured in the metrics file; the cross-language subset measurement is vacuously empty on this corpus. None touches an adjudicated quantity; all were disclosed by the independent validation, not by the authors — recorded as such.
3. **P3 (the Spanish-morphology mechanism) is untested:** the public corpus carries no Spanish-tagged documents, the registered power guard fired, and the mechanism claim awaits a successor on the private estate corpus where the subset exists.
4. **RRF  $k=60$  was never swept** (conceded to [3]); the retirement is of this configuration on this corpus class, not of fusion in general.

5. **Two accepted questions share an identical gold string** across different documents (symmetric across arms; worst-case 2/115 on P5).
6. **An adjudicator invocation occurred while records were incomplete**; its output was discarded and the final adjudication ran on complete, verified-unique records — but the incident is not falsifiable from the bundle alone, and successor designs must make the analyzer assert record completeness before computing.
7. **The absolute P5 accuracies are policy-bound** (hydrate-all) and not comparable to the parent’s agentic-loop numbers.

## 9. What this changes, and what comes next

The framework this program instruments shipped fused as its default locator. That default is now evidence-contradicted on this corpus class and retires per its own pre-registered ablation registry; the redesign (BM25-primary, with the exact leg as a fallback or rank-aware signal) is a dated design decision outside this study, and any performance claim for it requires a fresh registration. Registered successors: the fusion redesign; the Spanish-mechanism test where its subset exists; and harness rules adopted from this study’s own defects — gold-string uniqueness as a generation gate, rank lists recorded to fusion depth, completeness assertions in adjudicators.

The parent study showed an authored index mis-routing an agent that could have searched. This study shows the search it should have used was itself leaving a fifth of the corpus unreachable — and that most of that gap closes with ranking, while the obvious way to *combine* retrievers made things worse and was caught only because the claim was staked before the data existed. The gauge, again, is the deliverable.

## Availability

The replication bundle (doi:10.5281/zenodo.21969901) contains the frozen harness and analyzers, the gated question set with its generation log and candidate cache, all run records (localization and downstream, both arms), the adjudication output, the corpus manifest, the registration and correction lineage, and the two validation session prompts, returns, and disposition. The parent corpus snapshot is published with the parent deposit (doi:10.5281/zenodo.21960138). GitHub mirror: [github.com/mentu-ai/agent-memory-allocation](https://github.com/mentu-ai/agent-memory-allocation) (c36-locator-bakeoff/).

## References

Identifiers were live-verified 2026-08-16; author lists are given only where independently verified.

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